

Lab 1 – Product Description

Joshua Adkins

Old Dominion University

CS411W, Spring 2026

Professor S. Hosni

January 31, 2026

Version 1

Table of Contents

1	Introduction.....	3
2	Product Description	3
2.1	Key Product Features and Capabilities	5
2.2	Major Components (Hardware/Software).....	5
3	Identification of Case Study.....	6
5	Glossary	9
6	References.....	10

1 Introduction

Food product recalls have become an increasingly frequent occurrence in the United States, with federal agencies issuing hundreds of recall notices annually to protect public health. The Food Safety and Inspection Service reported extensive recall activity in 2023, documenting cases affecting millions of pounds of food products distributed across the country (FSIS, 2024).

Despite the existence of robust monitoring systems maintained by the Food and Drug Administration, the United States Department of Agriculture, and the Consumer Product Safety Commission, a critical gap persists between when recalls are issued and when affected consumers become aware of them. This delay can have severe consequences, as demonstrated by recent foodborne illness outbreaks such as the 2025 listeria outbreak linked to prepared pasta meals investigated by the Centers for Disease Control and Prevention. In some cases, the failure to effectively communicate recalls has resulted not only in widespread illness but also in criminal prosecution, as seen in the Conagra subsidiary case involving salmonella-contaminated peanut butter (US Justice Department).

The problems consumers face with recall awareness are considerable. While recall information is publicly available through government databases and press releases, the current system relies on consumers to actively seek out this information or notice it through news coverage. This passive approach creates four distinct deficiencies. First, an awareness gap exists where consumers simply do not know that products in their homes have been recalled. Second, an engagement gap arises because traditional notification methods fail to capture consumer attention or motivate timely action. Third, a customization gap prevents individuals from filtering recall information to match their specific purchases, dietary restrictions, or health vulnerabilities. Finally, an urgency gap means that even when consumers eventually learn about recalls, the notification often occurs

after they have already consumed the dangerous product. These gaps affect everyone, but they pose risks to vulnerable populations including parents of children with food allergies, immunocompromised individuals, and elderly persons who may have difficulty navigating multiple government websites to stay informed.

Because of these issues, effective recall notification and personalized product monitoring are essential. Currently, there is a lack of tools available that support both timely communication and the automated matching of recall data to individual purchase histories. Research presented at the 2025 Food Safety Summit emphasizes that recall modernization initiatives must prioritize consumer-centric design, acknowledging that traditional broadcast approaches cannot adequately serve the diverse needs of the population (Hallman, 2025). An effective solution must aggregate verified data from multiple federal agencies, deliver notifications through channels consumers use, and personalize alerts based on individual inventories and health profiles. SpoilerAlert is a mobile and web application designed to improve recall awareness and facilitate effective exchange of information between government recall systems and consumers. This application provides users with tools that improve efficiency and accessibility to information, thereby allowing consumers to protect themselves and their families; security, as SpoilerAlert is a platform for managing personal product inventories with appropriate privacy protections; immediacy, so consumers can receive real-time, reliable notifications when recalls affect their purchases; and clarity, because users will have the tools to receive actionable updates about specific products they own.

2 Product Description

SpoilerAlert is a cross-platform web and mobile application designed for use by consumers who want to stay informed about product recalls affecting items in their possession. It utilizes a personalized inventory system to maximize the efficiency of recall notifications by matching government recall data against products that users have purchased. The application aggregates recall information from the FDA, USDA, and CPSC databases and cross-references this data against user-specific inventories to deliver timely, relevant alerts through email and mobile push notifications.

2.1 Key Product Features and Capabilities

The key features of SpoilerAlert are its enhanced inventory management capabilities such as barcode scanning and receipt processing; its real-time recall monitoring that continuously checks FDA, USDA, and CPSC databases; and its personalized notification system featuring email and mobile push notifications. The barcode scanning functionality allows users to quickly add individual products to their inventory by scanning UPC codes with their smartphone cameras. The receipt scanning feature processes entire shopping receipts to log multiple purchases simultaneously, minimizing the effort required to maintain comprehensive inventory records. The application also provides customizable filtering options to allow users to specify allergens of concern, geographic preferences, and product categories such as food, cosmetics, and medicine. SpoilerAlerts cloud-based architecture ensures that inventory updates and recall alerts synchronize across all user devices, so a purchase logged on a smartphone immediately becomes available on the user's tablet or computer.

Consumer users can effectively monitor their product safety by having all related recall information and inventory records accessible on both their smartphones and desktop browsers. SpoilerAlert helps them protect their families by sending push notifications and email alerts when products in their inventory are recalled. The application provides a personalized dashboard that summarizes all active recalls affecting stored products, giving users an at-a-glance view of potential risks. Users can view real-time updates to recall information so that they are always aware of current safety issues. They can manage their product inventory, schedule notifications for products they plan to purchase, and receive alerts about recalls matching their specified allergens even if they have not yet scanned those products. The search and filter functionality allows users to proactively check for recalls by product name, brand, or UPC code.

SpoilerAlert benefits users by giving them more control over their family's safety. Users can view comprehensive recall information aggregated from multiple government sources without having to visit separate FDA, USDA, and CPSC websites. They can build personalized inventories through convenient scanning features rather than manually cross-referencing purchases against recall databases. Users will have access to immediate notifications that alert them to dangers before consumption or use occurs, fundamentally changing the recall awareness paradigm from reactive information seeking to proactive protection.

2.2 Major Components (Hardware/Software)

The mobile applications are compatible with both Android and iOS platforms, and any PC or other device with an internet connection can access the web interface. Users have access to both the mobile and web applications and mainly use, after being authenticated, the various inventory management, barcode scanning, and notification management features. The recall updates they

receive come from automated server processes that continuously monitor government API endpoints and cross-reference new recall data against the user database to identify affected individuals.

The major components of the application are structured according to the Model-Flow-Control-Display architecture. The model layer manages data structures including user accounts, product inventories, recall records, and notification preferences. The flow layer orchestrates navigation between application screens such as the dashboard, inventory management, barcode scanning, and settings interfaces. The control layer handles user interactions, coordinates API communications with both government databases and backend servers and manages local data persistence and synchronization. The display layer presents information through responsive interfaces optimized for mobile and web platforms, ensuring consistent user experiences across devices.

SpoilerAlert incorporates several software components and services. The mobile applications are built using cross-platform development frameworks to ensure optimal performance and access to device hardware features such as cameras for barcode and receipt scanning. For the server infrastructure, cloud-based services host the API framework and provide scalable computing resources to handle continuous recall monitoring and user notifications. The backend services integrate with the FDA OpenFDA API for food and pharmaceutical recalls, the USDA FSIS API for meat and poultry products, the CPSC API for consumer product safety information, and commercial barcode lookup services that translate UPC codes into detailed product information. Push notification services deliver real-time alerts to mobile devices, while email integration handles notification delivery to users who prefer email alerts. For the database, cloud-based storage solutions host user accounts, inventory records, and cached recall data, with encryption

ensuring privacy and security of user information. The system also includes manual data enrichment processes where team members research and link UPC codes to recalls when government announcements lack specific product identification information.

3 Glossary

Allergen: A substance, typically a protein, that causes an allergic reaction in certain individuals. Common food allergens include peanuts, tree nuts, milk, eggs, wheat, soy, fish, and shellfish.

Application Programming Interface (API): A set of protocols and tools that allows different software applications to communicate with each other. SpoilerAlert uses government APIs to retrieve recall data from FDA, USDA, and CPSC databases.

Consumer Product Safety Commission (CPSC): A U.S. government agency responsible for protecting the public from unreasonable risks of injury or death associated with consumer products. The CPSC issues recalls for various consumer goods including household items, toys, and electronics.

Contamination: The presence of harmful substances in food or products that make them unsafe for consumption or use. Common contaminants include bacteria such as salmonella and listeria, foreign materials, or undeclared allergens.

Food and Drug Administration (FDA): A U.S. government agency responsible for protecting public health by ensuring the safety and efficacy of food products, pharmaceuticals, cosmetics, and medical devices. The FDA issues recalls for products under its jurisdiction.

Immunocompromised: A condition in which an individual's immune system is weakened or impaired, making them more susceptible to infections and foodborne illnesses. This includes individuals undergoing cancer treatment, organ transplant recipients, HIV patients, and elderly persons.

Model-Flow-Control-Display (MFCD): A software architecture pattern that separates an application into four distinct components: Model (data structures), Flow (navigation logic), Control (user interaction handling), and Display (user interface presentation). This architecture promotes organized code structure and maintainability.

Product Recall: The action of removing or correcting products that violate laws administered by government agencies. Recalls may be voluntary by manufacturers or mandated by regulatory agencies when products pose safety risks to consumers.

Push Notification: An automated message sent from an application to a user's mobile device or web browser. Push notifications appear even when the application is not actively open, making them effective for delivering time-sensitive recall alerts.

Receipt Scanning: A feature that uses optical character recognition technology to photograph and automatically parse grocery receipts, extracting product information to populate a user's inventory without manual data entry.

Universal Product Code (UPC): A standardized barcode used to identify products in retail environments. UPCs consist of a series of numbers and bars that encode product information, enabling SpoilerAlert to match scanned products against recall databases.

United States Department of Agriculture (USDA): A U.S. government agency responsible for developing and executing federal laws related to farming, forestry, rural economic development, and food. The USDA's Food Safety and Inspection Service (FSIS) issues recalls for meat, poultry, and processed egg products.

4 References

- Centers for Disease Control and Prevention. (n.d.). *Listeria outbreak linked to prepared pasta meals*. <https://www.cdc.gov/listeria/outbreaks/chicken-fettuccine-alfredo-06-25/index.html>
- Food Safety and Inspection Service. (2024, February 28). *Summary of recall and PHA cases in calendar year 2023*. U.S. Department of Agriculture. <https://www.fsis.usda.gov/food-safety/recalls-public-health-alerts/annual-recall-summaries/summary-recall-and-pha-cases-0>
- Hallman, W. (2025). Recall modernization initiatives with the consumer in mind. In *2025 Food Safety Summit*. https://www.food-safety.com/ext/resources/FSS_Event/Presentations/2025/S7---Combined-Slides.pdf
- Schildhouse, J. (2024, December 20). Why are there so many food recalls right now? *Taste of Home*. <https://www.tasteofhome.com/article/why-are-there-so-many-food-recalls-right-now/>
- U.S. Department of Justice. (n.d.). Conagra subsidiary sentenced in connection with outbreak of salmonella poisoning related to peanut butter. <https://www.justice.gov/archives/opa/pr/conagra-subsiidiary-sentenced-connection-outbreak-salmonella-poisoning-related-peanut-butter>